

# Privacy-preserving multidimensional big data analytics models, methods and techniques: A comprehensive survey

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## ABSTRACT

The topic of *privacy-preserving big data analytics* is gaining momentum now, thanks to the plethora of modern application scenarios where it can be successfully applied. These range from smart city applications to e-government, from social networks to graph-like analysis tools, from logistic transport systems to business intelligence systems, and so forth. *Healthcare systems*, for instance, are turning out to be one of the major applicative settings where privacy-preserving big data analytics tools are extensively applied, due to the specific requirements dictated by the context (i.e., protecting individual privacy). *Multidimensional big data analytics methods* are a special case of big data analytics methods where the emphasis is put on the usage of *multidimensional metaphors*, which had a fortunate experience in *OnLine Analytical Processing* and *Business Intelligence* systems. One step forward, *privacy-preserving multidimensional big data analytics*, is thus of extreme interest to the research community because it combines the intrinsic privacy-preserving problem with the emerging multidimensional big data analytics methods. Inspired by these considerations, in this paper, we provide a comprehensive overview that focuses the attention on privacy-preserving multidimensional big data analytics models, methods, and techniques over big data by highlighting several characteristics of the various in-literature proposals and also providing a review of future research challenges in the investigated research domain.

## 1. Introduction

In the context of *big data analytics* (e.g., Berisha, Mëziu, and Shabani (2022), Kambatla, Kollias, Kumar, and Grama (2014) and Tsai, Lai, Chao, and Vasilakos (2015)), *multidimensional big data analytics* (e.g., Bochicchio, Cuzzocrea, and Vaira (2016), Cuzzocrea (2015, 2020, 2021, 2022a, 2022b) and Cuzzocrea and Bringas (2022)) plays a leading role, mostly due to the wide family of real-life applications where it can be successfully applied. Indeed, there is a long list of applicative settings, ranging from *big social data* (e.g., Erol, Aksoy, and Sagioglu (2023)) to *big sensor data* (e.g., Ren, Wang, Zhang, and Zhang (2023)), from *big graph data* (e.g., Yuan et al. (2023)) to *big energy data* (e.g., Chen and Zhu (2023)), and so forth, where the intrinsic nature of *multidimensional abstractions* (Cuzzocrea, 2015), provides a clear advantage over classical, e.g., SQL-based interfaces, methodologies via an undoubted expressive power that has no equal in the well-known class of analytics methodologies. As a result, several data domains can be easily modeled in terms of *dimensions*, *hierarchies*, *levels*, and *measures* (Gray et al., 1997), and powerful analytical tools can be built upon them.

*What multidimensional big data analytics can do more than yet-classical big data analytics?* This is a critical question to be addressed, as it represents a clear revolution for actual big data research (e.g., Cuzzocrea, Song, and Davis (2011)). Indeed, traditional *OnLine Analytical Processing* (OLAP) and *Business Intelligence* (BI) systems (e.g., Chaudhuri, Dayal, and Narasayya (2011) and Gray et al. (1997)) have already proven to noticeably extend the expressive power of SQL-based data analytics tools, as it is well-known that, usually, *real-life datasets naturally expose a multidimensional and multi-level nature*. Therefore, multidimensional analytics tools are more suitable to extract “actionable knowledge” from these big datasets rather than traditional, adimensional approaches. Not by chance, this evidence has already been highlighted by recent studies in the area (e.g., Cuzzocrea et al. (2011)).

All the mentioned applicative settings expose a non-trivial research challenge, i.e., preserving the *privacy* of the target datasets. *Privacy-preserving big data analytics* (e.g., Oehmichen, Jain, Gadotti, and de Montjoye (2019) and Tran and Hu (2019)), is a keen topic in next-generation big data research, which has attracted the attention of a larger community of both academic and industrial researchers. The

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problem is represented by the issue of preserving the privacy of *sensitive* information kept in the target datasets. A clear example of such a scenario is represented in the *healthcare domain* (e.g., Chauhan, Kaur, and Chang (2021), Coronato and Cuzzocrea (2022) and Mavriki and Karyda (2020)). Indeed, *personal* information, such as name, surname, age, address, and so forth, cannot be exposed in the derived big data analytics due, for instance, to legal constraints e.g., GDPR (*Global Data Protection Regulation* – (Commission, 2023) – in force into European Countries). Methodology-wise, this translates into the necessity to preserve the privacy of *identifying attributes* through big data analytics processes.

As a consequence, we argue that the problem is consequently extended to the more probing case of dealing with multidimensional big data analytics issues. Intuitively enough, due to the (high) multidimensionality and the (high) complexity of target big datasets (e.g., Cuzzocrea et al. (2011)), then ensuring the privacy of the overall multidimensional big data analytics process represents a challenge to be investigated by next-generation efforts in big data research.

In this context, *privacy-preserving multidimensional big data analytics* (e.g., Cuzzocrea (2021)) is a special branch of the main privacy-preserving big data analytics area, where the focus evolves around a combination of multidimensional big data analytics methodologies and privacy-preservation paradigms. This research context predicates preserving the privacy of a sensitive range of data while supporting multidimensional big data analytics tasks. This is a very ambitious goal that turns out to be a real “call to arms” for actual and future research efforts. It is worth mentioning that this surrounding movement is just stirred-up by the recent impressive explosion of *modern Cloud and big data processing platforms* (e.g., Zhang, Qi et al. (2022)).

Several traditional literature approaches have, for instance, focused the attention on the related *privacy OLAP* context (e.g., Agrawal, Srikant, and Thomas (2005)), where the problem consists of, given a multidimensional OLAP data cube, computing the privacy-preserving version of the data cube, according to a given criterion (e.g., invariance of range query answers – (Sung, Liu, Xiong, & Ng, 2006), trade-offing privacy and accuracy of privacy-preserving output data cubes – (Cuzzocrea & Saccà, 2010), and so forth). After computing the privacy-preserving OLAP data cube, then multidimensional big data analytics processes are supposed to be privacy-preserving as well. This is, indeed, one of the possible approaches, but, according to another vision we argue in this paper, the privacy-preservation should *also* concern with the intrinsic multidimensional big data analytics process model, thus involving the process model itself, beyond to the target (big) datasets.

Under a larger vision, the paradigm above thus realizes a *framework-oriented* approach, where the *privacy design* concerns the whole big multidimensional analytics process as well as the target big dataset. It should be noted that the latter one is an innovative vision over actual state-of-the-art proposals that, contrary to this, address the two combined issues separately (e.g., Agrawal et al. (2005) and Sung et al. (2006)).

This paper lies in this so-delineated research context. Looking at the active literature, we recognize that few proposals can be specifically co-located in this setting. Therefore, there is a *clear* need for systematically establishing the privacy-preserving multidimensional big data analytics research area as well as highlighting definitions, properties, benefits, and limitations.

Based on the above motivations, this paper proposes a detailed survey of models, methods, and techniques falling in the privacy-preserving multidimensional big data analytics over big data discipline and provides a critical review of the benefits and limitations of state-of-the-art proposals. The paper also provides a milestone-inspired overview of the main future research directions in the investigated field.

## 1.1. Research questions

The intrinsic goal of this paper consists in replying to the following research questions:

- **RQ1.** *Which are the main pillars of the research corpus in privacy-preserving multidimensional big data analytics?*  
While a plethora of proposals exist in the context of privacy-preserving multidimensional big data analytics, one of the main goal of this research consists in defining a systematic methodology for identifying the pillars of this scientific context, and then classifying the relevant state-of-the-art proposals with respect to these pillars. Pillars are defined as the most relevant and characterizing research topics in the investigated area.
- **RQ2.** *Which are the most-relevant and emerging proposals in the context of privacy-preserving multidimensional big data analytics, and how to well-summarize them?*  
“Putting order” within a very wide corpus of research proposals falling in the emerging context of privacy-preserving multidimensional big data analytics and really selecting the most-relevant and emerging ones is not a trivial exercise. A first-class goal of this research is to fulfill this important gap, thus contributing to the research community.
- **RQ3.** *Which are the most-noticeable and emerging challenges and future research directions in the context of privacy-preserving multidimensional big data analytics?*  
A comprehensive survey not only must analyze, classify and characterize relevant research proposals of the investigate scientific context, but also challenges and future research directions must be identified and discussed, in order to support future efforts in the area. The latter is another-yet-prominent goal of our research, since the results of our study may inspire and drive other research experiences that focus the attention on privacy-preserving multidimensional big data analytics models and techniques.

## 1.2. Structure of the actual study

Given the difficulty and complexity of the investigated scientific field, this study introduces seven specialized topics, called *pillars*, each one falling under the main area of privacy-preserving multidimensional big data analytics, but with a particular emphasis on certain facets of the main scientific and methodological scope.

In our study, we identify the following main specialized topics:

- *Multidimensional Big Data Analytics;*
- *Privacy of High-Dimensional Data;*
- *Privacy OLAP;*
- *Privacy of Big Multidimensional Data in Emerging Application Scenarios;*
- *Privacy via Anonymization in Big Data Analytics Systems;*
- *Privacy via Multidimensional Anonymization in Big Data Analytics Systems;*
- *Architectures and Platforms for Advanced Privacy-Preserving Analytics on the Clouds.*

These topics are the main focus of our study, and for each of them, we provide a detailed description. It should be noted that, taken as singular, topics represent a proper (sub-)scientific area, but our main goal is that of constructing and assessing a “unified” research perspective where they cooperatively contribute to the emerging research challenge represented by the privacy-preserving multidimensional big data analytics paradigm. This validates the goodness of our proposed study.

### 1.3. Summary of contributions

In this paper, we present and deeply discuss a comprehensive survey on privacy-preserving multidimensional big data analytics models, methods, and techniques over big data.

Overall, in this paper, we make the following contributions:

- we initiate by identifying the seven specialized topics and pillars within the context of privacy-preserving multidimensional big data analytics;
- we also provide a very detailed state-of-the-art analysis on the basis of the previous pillars, which allow us to provide a comprehensive vision of the most relevant and pertinent work related to our research;
- we define a set and various methods and techniques related to this investigated context that will be used to perform a classification and categorization of the relevant literature research works;
- we summarize in a table the selected references based on their membership to the described set of properties;
- finally, we illustrate a review of future research challenges in this investigated domain.

### 1.4. Paper organization

The remaining part of this paper is organized as follows. In Section 2, we provide the formal background on the multidimensional big data analytics discipline. Section 3 introduces the seven pillars that are the basis of our study. In Section 4, we focus the attention on a deep analysis of state-of-the-art contributions in this area. Section 5 introduces the main concepts concerning which we classify the state-of-the-art papers we select from active literature. In Section 6, we provide the summary of our survey study to better focus on the effective outcome of our research. Section 7 reports on challenges and future research directions in the investigated scientific context. Finally, in Section 8, we outline the conclusions and future work of our research.

## 2. Background on multidimensional big data analytics

Multidimensional big data analytics (e.g., [Bochicchio et al. \(2016\)](#), [Cuzzocrea \(2015, 2020, 2021, 2022a, 2022b\)](#) and [Cuzzocrea and Bringas \(2022\)](#)) refers to a new line of research where the emphasis lies on the *multidimensionality of the analytical model*, which may or may not marry with the multidimensional nature of target big datasets.

The term has been proposed in 2011 ([Cuzzocrea et al., 2011](#)) as a new challenge for next-generation big data research, being preceded by some works where the standard data representation and mining tasks have been, firstly, combined with multidimensional modeling concepts (e.g., [Gray et al. \(1997\)](#)). Over time, the interest from the research community has stirred up, mostly towed by an astonishing growth of technology like social network platforms and Cloud computing, from one side, and applications like smart city and bio-informatics, from another side.

Multidimensional big data analytics extends the power of big data analytics by injecting new concepts in the analytical model, mostly inherited from classical OLAP and BI experiences, thanks to the model-axioms recognized under the terms *dimensions* and *measures*. Given a target big data domain  $D$ , the first dimensions are identified. Dimensions are model attributes according to which the main big data analytics process is conceived. Then, measures are identified. Measures are model attributes according to which the main big data analytics process is executed.

Just to give a first, toy example, we consider the well-understood *healthcare setting*, for instance as related to the COVID-19 outbreak. Here, data analysts may be interested to analyze the number of *infected people* (i.e., the measure of the target multidimensional big data analytics process) in dependence on the *zone, time, user-profile*,

*medical-history, comorbid-factors*, and so forth (i.e., the dimensions of the target multidimensional big data analytics process). All these concepts convey to define a clear multidimensional model against which not only classical OLAP operations can be carried on (e.g., [Gray et al. \(1997\)](#)), such as *pivoting, drill-down, roll-up*, and so forth, but, most importantly, data-mining algorithms, such as *frequent pattern* (e.g., [Lee et al. \(2013\)](#)), and machine-learning methods, such as *regression* (e.g., [Nedjar, Casali, Cicchetti, and Lakhali \(2009\)](#)), can be applied on, thus magnifying the *expressive power* of the underlying process.

[Fig. 1](#) provides a reference architecture for the case study related to the actual healthcare setting.

As shown in [Fig. 1](#), several layers can be identified in the reference architecture. In the next, we describe these layers:

1. *Big Data Domain*;
2. *Multidimensional Modeling*;
3. *Multidimensional (Big) Data Model*;
4. *Google Cloud Platform*;
5. *Multidimensional Big Data Analytics Tasks*;
6. *Multidimensional Big Data Analysis*.

## 3. The seven pillars of privacy-preserving multidimensional big data analytics models, methods and techniques over big data

In this Section, we introduce the main pillars, under which we select several reference research works to describe and detail. The specialized topics are introduced to help frame the scope of this work.

### 3.1. The multidimensional big data analytics pillar

*Multidimensional big data analytics* are decision-making-enabling knowledge that is provided by multidimensional big data analysis over multidimensional data models defined in multidimensional big data storage systems, such as those leveraging data warehouse capabilities (e.g., OLAP servers). Multidimensional big data analysis is a task involving a customized and oriented scrutinizing of the whole of data stored in an adequate multidimensional model such as OLAP. Usually, multidimensional big data analysis is enabled by targeting specific dimensions of data and inspecting (e.g., using an aggregation operation) at their confluence to derive useful knowledge for a specific application. Examples of multidimensional big data analytics-enabling systems are presented in [Ahsan, Teay, and Albarbar \(2020\)](#), [Guerrero-Prado, Alfonso-Morales, and Bravo \(2021\)](#) and [Sinha, Bhatnagar, and Bansal \(2017\)](#).

### 3.2. The privacy of high-dimensional data pillar

*Privacy of high-dimensional data* is an open challenge that poses burdening defiance to actual scientific knowledge and results on the topic. Indeed, many data privacy techniques fall short in dealing with a high number of attributes when privacy-protecting “high-dimensional” data. It has been established that most existing privacy techniques cannot handle high-dimensional data properly and efficiently ([Aggarwal, 2007](#)). On the other hand, applying existing privacy techniques to high-dimensional data is associated with a degradation of the resulting data quality used for data analysis ([Fung et al., 2012](#)). In [Aggarwal \(2005\)](#), it has been noted that techniques such as *k-anonymity* depend on spatial locality (measured through a distance metric) of data points to perform the anonymization of the data, however, spatial locality becomes distorted in the case of high-dimensional spaces ([Aggarwal, Hinneburg, & Keim, 2001](#)). A feature selection solution is not very helpful since it results in discarding a number of features, a process that may negatively impact data utility. However, a promising direction worth mentioning is described in [Zakerzadeh, Aggarwal, and Barker \(2014\)](#), which suggests employing a vertical fragmentation approach to divide data according to their attributes and to anonymize each part independently.

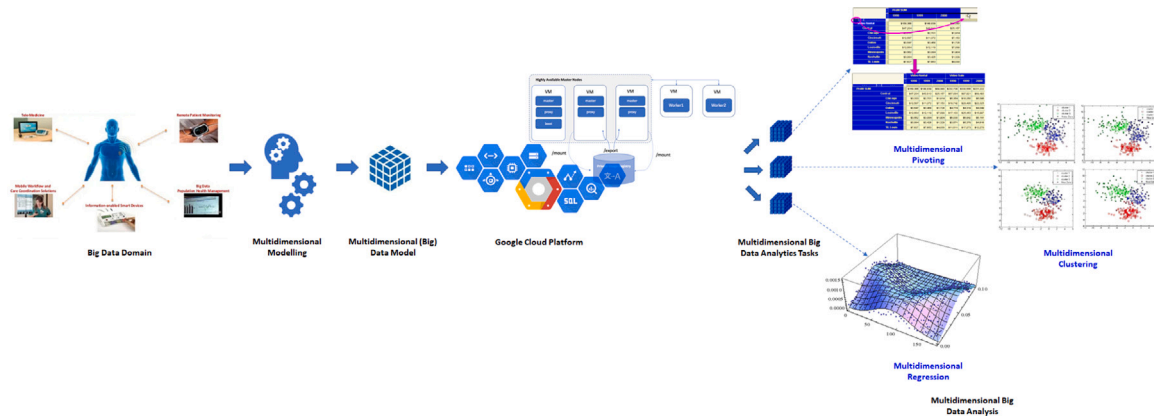


Fig. 1. Case study: Supporting multidimensional big data analytics over healthcare settings: A reference architecture.

### 3.3. The privacy OLAP pillar

OLAP is a decision-support method that enables precious knowledge to be mined through the exploration of a multidimensional data structure. Still, in many cases, the exploration should be performed in a private way. Indeed, for instance, sensitive information could be exposed to serve malicious intentions, and the need for privacy is thereby obvious. OLAP privacy might be implemented in several ways: for example, [Priebe and Pernul \(2000\)](#) presents an access control approach over OLAP cubes that limits the access to untrusted users by concealing slices of data before granting access to the cube. The concealment is done according to two ways: *views way* or *rules way*. Views way restricts the display of certain sensitive dimensions, while rules way limits the access to sensitive data for certain users and not to others. Another example ([Agrawal et al., 2005](#)) describes a method to perturb OLAP-queried client data (through retention replacement schemes that select the target data to randomize) before their actual integration into the server. The data are then reconstructed using a matrix inversion technique or an iterative Bayesian technique before applying the aggregations. Another example ([Cuzzocrea, Russo, & Saccà, 2008](#)) proposes a sampling method approach to protect OLAP cubes.

### 3.4. The privacy of big multidimensional data in emerging application scenarios pillar

Emerging technologies are dictating stricter than ever conditions on multidimensional big data processes, methods, and environments. In particular, privacy techniques over big multidimensional data are required to evolve into more effective, more efficient, more flexible, and more scalable. Indeed, given the new ways data are stored, processed, and analyzed, more comprehensive privacy techniques should be defined to cope with privacy challenging technological breakthroughs. In the case of smart healthcare, nowadays, *Electronic Health Records* (EHR) are considered as the new reference data source, enabling, for example, considerable process improvement in the healthcare domain, among many other benefits ([Chao, Hu, Ung, & Cai, 2012](#); [Wikström et al., 2019](#)). In [Wu, Wang, Ning, and Zhu \(2022\)](#), EHR sharing is privacy protected through an innovative system leveraging blockchain-enabled dynamic access control combined with local differential privacy. Another example explains how it is possible to conceive a privacy-preserving mobility data framework “by-design” ([Monreale, Rinzivillo, Pratesi, Giannotti, & Pedreschi, 2014](#)). Yet, another example describes a method to anonymize smart metering data ([Efthymiou & Kalogridis, 2010](#)). The approach consists of distinguishing high-frequency (used to evaluate regular user’s power consumption) and low-frequency smart data (used to generate billings to users each month) and performing the anonymization of solely a *user ID* associated with their generated high-frequency metering data to enable secure smart metering data exchange between components of the smart metering installations. The anonymization of the such user ID is based on an escrow service.

### 3.5. The privacy via anonymization in big data analytics systems pillar

Big data analytics is useful in decision-making ([Elgendy & Elragal, 2016](#)). As this is becoming a requirement, making big data analytic systems more responsible for privacy entails embedding a privacy module that takes good care of concealing sensitive data when applying analytical procedures. In this sense, [Lin et al. \(2016\)](#) describes an anonymization method in the context of a *Body Area Network* system, where wearable sensor devices are responsible for collecting electrocardiogram (ECG) data. The method is about adding noise to ECG dataset features-based on their order of importance (established in a data pre-process phase) to reach a well-defined differential privacy interference threshold. The interference threshold is updated on the fly as new data is measured from the sensory devices and takes into account feature importance. The interference threshold is responsible for guaranteeing a certain level of privacy for the data. Another example of such system is described in [Zhang, Chen, Yu, and Deng \(2019\)](#), where the proposed system utilizes *Paillier Homomorphic Cryptosystem* to prevent information leakage at the network channels level in a federated learning system context.

### 3.6. The privacy via multidimensional anonymization in big data analytics systems pillar

The multidimensional constraint of data is a crucial aspect to consider in the context of a privacy-preserving big data analytics system. The multidimensionality aspect of data adds, yet another, *privacy-hampering* obstacle, application-independent, to big data analytics. In an attempt to elude the mentioned problems, one example proposed in [Zhang, Huang et al. \(2022\)](#) uses a variant of the *BGN encryption* algorithm to enable the multidimensional encrypted data aggregation needed to collect data from the data sources at the edge network into a control center, from which analytics could be derived privately.

### 3.7. The architectures and platforms for advanced privacy-preserving analytics on the cloud pillar

Cloud services are needed more than ever for a plethora of intensive data-related processes, such as big data analytics. The extremely high volume of data generated through new technologies dictates the need for more resources to accomplish data-related tasks. In particular, privacy-preserving data analytics requires copious computing and storage resources to perform adequately under tighter data volume-related constraints. An example of solution to the previously mentioned issues is described in [Poddar, Ananthanarayanan, Setty, Volos, and Popa \(2020\)](#). The detailed solution consists of a cloud-based video analytics as a service solution that protects video streams from side-channel attacks as well as from direct leakage of private video content. Another

example (Xu et al., 2019) presents a solution that uses differential privacy techniques to enable data sharing and analytics functionalities as cloud services. Bondel, Garrido, Baumer, and Matthes (2020) argues that the best-to-use anonymization methods in a cloud-based context are the de-identification approaches such as pseudonymization, generalization, suppression, or also data masking. They exclude the suitability of other anonymization-enabling approaches such as *homomorphic encryption* or also partial encryption in combination with trusted client/hybrid clouds in a cloud context for several stated reasons.

#### 4. Analysis of state-of-the-art: A comprehensive vision

Based on the introduced seven pillars of privacy-preserving multidimensional big data analytics models, methods, and techniques over big data provided in Section 3, here in this Section we specifically focus the attention on some specific contributions that, according to our study, represent relevant research results worth to be mentioned. Still, we make use of the conceptual classification of our pillars, which constitutes one of the main contributions of our research.

##### 4.1. Multidimensional big data analytics

In this Section, we report on some of the most relevant related work within the context of multidimensional big data analytics.

Bochicchio et al. (2016) proposes a framework for analyzing multidimensional data by applying clustering and classification algorithms to multidimensional views of input clinical data. The approach aims at finding anomalies in gestated human fetuses for early detection of diseases as well as for early spotting of defective fetuses. The approach summarizes in building fetal growth curves (through analyzing fetus data with OLAP using *Dimensional Fact Models*), which would serve as benchmarks for comparing fetuses' growth and thereby spot anomalies. The main assumption consists of the fact that normal fetuses share similar baseline growth characteristics if they have the same gestational age, similar genetic properties (e.g., ethnicity), and have grown in similar environmental conditions (e.g., food). Baseline fetuses' growth curves ease the definition of groups of fetuses called *Homogeneous Patient Groups* (HPG). The membership of fetuses to one group of HPG will inform about their health (*wellbeing* and *pathologies*). Multidimensional views are created from target data, after a classification algorithm is applied to the data to find the closest HPG group for a certain fetal patient.

In Tanaglia et al. (2021), authors propose a tool for analyzing multidimensional healthcare biological treatment-related data and monitoring their administration to the patients. They propose a micro-services-based tool where different components are deployed using container virtualization (e.g., Docker) to analyze ROLAP-based data stored in the PostgreSQL database. They propose a dimensional fact model (for the data mart) for analyzing biological drug administrations for the patients.

##### 4.2. Privacy of high-dimensional data

In this Section, we report on some of the most relevant related work within the context of the second pillar, privacy of high-dimensional data.

Ghinita, Kalnis, and Tao (2011) present a technique to capture the correlation of the data while anonymizing them, accounting for the maximization of data utility. The example presented in this paper illustrates how the correlation matrix is generated, as follows: starting from "transactional data" made by some customers, the general matrix is built by placing a 1 when an item is within a transaction, a 0 if not. Deliberately, sensitive attributes are associated with sensitive items in order to ease the processing. They first start by converting the general matrix of the data into a band matrix through the Cuthill-McKee

algorithm (Cuthill & McKee, 1969). Next, they use it to create formations of sensitive transactions through their proposed greedy algorithm, CAHD. CAHD works as follows: an iteration over a sensitive transaction consists of the grouping of  $p$  nearby (input privacy  $p$  degree) sensitive transactions together based on a notion of non-conflict attributes (two conflicting sensitive transactions have a common sensitive value) and through selecting together only transactions that have the closest number of common Quasi-ID. During the process of group formation, an histogram is maintained to guarantee that the constraint of privacy is satisfied. The algorithm halts when no more ungrouped sensitive transactions remain, or when no additional groups can be formed.

In Abdalaal, Nergiz, and Saygin (2013), authors introduce an augmentation of  $l$ -diversity (*MSA-diversity*), which is an algorithm capable of protecting against what they refer to as, attacks using non-membership knowledge: the probability to infer the victim's record is bounded by  $1/(l-i)$  under  $i$  bits of non-membership knowledge (non-membership knowledge: the attacker knows  $i$  sensitive attribute values about the victim's equivalence class, information that would narrow the number of possible records related to the victim by shrinking the number of potential victim's records). The algorithm proceeds to the mentioned goal by, first, linearizing the multidimensional Quasi-ID attribute values (tuples) into an ordered one-dimensional list, then trying to construct SA-distinct (through defined MSA-diversity) groups from these tuples by creating associated matrices where no two tuples share a row or a column. The groups are then generalized or anatomized.

Prasser et al. (2016) presents a multidimensional anonymization technique where generalization transformations are applied iteratively to the target dataset. The transformations are set in a lattice's vertices, and the algorithm traverses the lattice accounting for the maximization of data utility and a certain defined data risk threshold (a transformation is validated and used only if it increases data utility and if data risk thresholds are met). To achieve its goal, the algorithm toggles between a greedy traversal approach and a Best-first search approach, indeed, if the lattice limit is reached, the algorithm performs a backtrack to be able to continue the traversal. If  $n$  steps were taken, the algorithm will also move to use the greedy algorithm, but here without performing any backtracking. While traversing the lattice, the algorithm maintains a queue where transformations are ordered according to data utility (descending order). Intuitively, the more the transformation preserves data utility, the less the associated data generalization.

##### 4.3. Privacy OLAP

In this Section, we explore the concept of privacy OLAP by focusing on presenting some relevant state-of-the-art work.

Cuzzocrea and Saccà (2012) proposes a three-step sampling method towards the anonymized analysis of data cubes, namely: storage space allocation for the samples extracted from the query workload ( $QWL$ ) to obtain the approximative data cube answer, the actual sampling of the data query workload step and a refinement phase included to enhance the results in terms of accuracy (of the answer) and privacy (of data), thereby securing a balanced trade-off between the two target metrics. The authors also define techniques to deal with numerous problematic aspects hindering an accurate and private query answer for the data cube. For example, they propose to look at a query over a dataset as a set of mini-queries defined over a set of "regions", which are the bounding spaces of the mentioned queries. From there, they represent a query as a set of regions that should comply with several conditions to enable a fine-grained and effective analysis while keeping the queried data safe from ill-intentioned purposes.

In Cuzzocrea and Bertino (2014), the authors propose a novel technique to anonymize data cubes in a distributed environment. They use CUR matrix decomposition (Drineas, Kannan, & Mahoney, 2006), a matrix decomposition method used to compute approximate representations of large matrices, to anonymize a 2D view of the distributed cubes. In the distributed scenario, they propose a *Secure Distributed OLAP* aggregation protocol (SDO) (Cuzzocrea & Bertino, 2010)

to anonymize the entirety of the data cube network. To stress the anonymization process achieved thanks to the CUR approximation, they perform an in-depth assessment of the reconstruction results by retracing the method first proposed by Agrawal et al. in *Retention Replacement Perturbation algorithm* (Agrawal et al., 2005). The idea is to quantify accurately enough the values (through setting reasonable and rigorous bounds) of the yielded range-SUM query answers on the reconstructed data. A comparison between the mentioned query results over both a constructed version and the initial one is also performed to confirm the effectiveness and efficiency of the method used (CUR decomposition method). They also demonstrate some useful mathematical properties involving what they name as full-differential privacy (a quantification of the difference between the initial and the transformed view) and marginal-differential privacy (a quantification of a partial difference between the initial and the transformed view on the queried attribute domain).

Kim, Lee, and Chung (2017) reveals the impact of three types of anonymization approaches, namely: global generalization, local generalization, and bucketization, over data utility and loss of information in anonymized data cubes. They pinpoint how the mentioned anonymization approaches could alter the performance of data cube analysis concerning three accounted-for metrics, namely: data cube size, cell overlaps, and information loss. Indeed, they argue that these metrics are unavoidably affected by the data alteration resulting from generalization or bucketization. In addition, they highlight how the considered anonymization techniques shape the final anonymized data cube. For instance, they explain that bucketization tends to increase the number of cells because buckets (sensitive values) are attributed to all of the associated Quasi-ID records in the anonymized data cube. Similarly, local generalization may lead to an increased number of cells in the anonymized final data cube, whereas global generalization leads to fewer cells. In addition, they provide an example of how local generalization leads to more overlapping cells and justify it by the fact that the method allows multiple recording rules. Concerning the information loss metric, they argue that all considered anonymization techniques lead to information loss more or less significantly. Besides, they propose a fourth metric named median relative error. The approach they use to emphasize the cited metric is through, first, by querying cuboids based on selected attribute values or ranges of attribute values, and second, by computing the relative error by executing each type of query (point and range) on different cuboids and calculating a rate consisting of the difference between the result of the query on the anonymized cuboid and the original dataset over the result obtained from the original data cube.

#### 4.4. Privacy of big multidimensional data in emerging application scenarios

In this Section, we review some related work on privacy of big multidimensional data in emerging application scenarios. We examine various studies and methodologies that address privacy concerns in contexts such as IoT, smart cities, healthcare data and so forth.

In Liu et al. (2022), data integrity and user privacy are the main privacy-preserving concerns of this research work. To reach their goal, the authors propose a technique where the data are, first, packed through the Chinese remainder theorem and, second, encrypted through an encryption technique that does not rely on an established public-key encryption mechanism (such as the Paillier method) but rather on an approach based on the user's message using a technique based on key negotiation between users. The mentioned technique relies on time-stamping each user's data to achieve its goal. According to the authors, the encryption technique guarantees resistance against the infamous network replay attack. Phases of the scheme proposed, most importantly, include a data reporting phase (SM), an anonymized data aggregation phase (GW), and finally a data parsing phase (CC).

Indeed, the developed system aims at collecting reports on electricity consumption from residential areas, including many residents (SM), to encrypt them and eventually forward them through gateways (GW) in an aggregated form. Their analysis is then done through *Control Centers* (CC). Given this setup, it goes without saying that the conveying of the data should be secure for the better of the users and the control center instance. The use case of this approach is related to smart grids, where these should be fed with data analysis in order to enhance power delivery strategies for the residents.

Peng, Luo, Wang, Khan, and He (2022) proposes to design an innovative homomorphic encoding algorithm based on the Chinese remainder theorem and the Paillier encryption. The encryption is performed after the IoT devices-based collection of the data step and before the aggregation of the same on the edge nodes and their sending to the control center for analysis safely and securely. The system model they envision is aimed at providing intelligent transportation services by analyzing device data. More into detail, the edge nodes will receive ciphertext from the IoT devices, will perform aggregations (Homomorphic Operations) on them, and will send the aggregated result to analysis, posterior to the decryption step, at the control center. The Chinese remainder theorem was used to transform the input message vectors into integer vectors, and the Paillier method was used to actually encrypt these vectors into ciphertext. Aggregations on top of encrypted data are possible since the homomorphic-based encoding of the data enables that.

Zheng et al. (2022) copes with privacy-preservation of both the initial data (residing in the healthcare center and then moved onto the cloud for ease of processing) and the query (e.g., range SUM query) submitted to the cloud by the users/doctors. They employ an *R*-tree to index the dataset and to facilitate the querying of the data through range SUM queries by defining an algorithm for the purpose. The main idea sums up in finding the intersecting data between a target portion of data (set in the traversed node) and the query intervals. The *R*-tree traversal is done in a depth-first way. More specifically, the top-bottom search of the tree continues only if the query intersects with the current node's data, and eventually leaves are added to the query answer if they satisfy the mentioned condition. The authors assimilate the boundaries of each of the node's data and the query as integers and reason on their binary vector version to compare them, providing, thereby, an anonymized version of the evoked algorithm through the comparison of the data and query boundaries. The comparison would inform about the intersection of the query with the data (and thereby on the set of query answers). The comparison of the integers is transformed into an equality test based on a function they define, which varies according to vector variables set from the vector version of the integers in question (through the conjugate and transpose of the original integer vectors). This comparison aims at concealing the intermediate operations (inequalities evaluation) results needed to obtain the resulting intersection between the query and the data. More specifically, they employ a homomorphic encoding-based data comparison technique (involving encryption of the vectors of data and the query and matrix-based calculations on the encoded versions of the vectors) that outputs the result of an equality test informing on the intersection between the original vectors.

In Liu, Gu, Shojafar, Alazab, and Liu (2023), they propose a privacy-preserving data aggregation solution implemented within a three-layer fog-based system aiming at analyzing data through user-specified queries while keeping data safe and secure. The authors propose an architecture formed of a *Service Server* (SS), *Fog Nodes* (FN), and *Data Terminals* (DT). FNs connect to SS, whereas FNs connect DTs (each DT collects data from, e.g., sensors through, e.g., Bluetooth). Several steps to querying the data are proposed, namely: system initialization, data collection, data blending, data aggregation, and finally the results reporting step.

#### 4.5. Privacy via anonymization in big data analytics systems

In this Section, we review some relevant related work on privacy via anonymization in big data analytics systems, by focusing on various anonymization techniques, such as  $k$ -anonymity,  $l$ -diversity, and  $t$ -closeness, and their application in preserving sensitive information within large datasets.

In [Mohammed, Fung, Hung, and Lee \(2010\)](#), the authors depart from a reasonable, privacy-breaching constraint assumption that most data records piracy acts are enabled through the knowledge of the attribute values corresponding to only a portion of the entire QID set (full attribute set of the quasi-identifiers of the dataset). Based on that, they propose a utility-aware privacy method named *LKC-privacy*. Namely, this method suggests that for each tuple of QID attributes of size lower than  $L$  included in the full set of QID, at least  $K$  records should share the same attribute values (guaranteeing  $K$ -anonymity). In addition, the probability of inferring a sensitive value within the equivalence classes should not exceed  $C\%$  (guaranteeing  $C$ -de-association). They propose two types of scenarios where anonymization of surgery and blood transfusion data are performed: a centralized approach, where a *Central Government Health Agency* (CGHA) anonymizes the data in *LKC-privacy* fashion for the recipient (Red Cross Blood Transfusion Service) delivery, and a distributed approach, where the data are disjointly anonymized at each hospital performing the blood transfusion to the patients, skipping thereby the need to go through CGHA.

[Onesimu, Karthikeyan, and Sei \(2021\)](#) proposes a  $k$ -anonymization-based approach of IoT health-related measures collected from patient-embedded devices such as medical sensors. The considered architecture of the system supposes that collected data (from patient's devices) are to be sent over a WAN, Internet, or Mobile network for healthcare services to be analyzed and to eventually serve as a baseline for the system to enable recommended treatments for the patients. The authors assimilate the system architecture to a client-server one, where data transits in between. The anonymization task occurs within the network, where an adversarial eavesdropper may interfere with the data transfer to an untrusted server. The authors propose to anonymize the data through a clustering-based approach using the  $\alpha$ -de-association technique at data aggregating nodes on the network. Initially, each tuple is considered a cluster, after which a representative vector is elected to compare distances between the clusters and generate a dissimilarity matrix. Clusters are grouped according to the similarity between the tuples. More specifically, cluster dissimilarities are used to evaluate the distance between the cluster pairs.

[Lawrance and Jesudhasan \(2021\)](#) defines a MapReduce-based clustering anonymization algorithm. The algorithm makes use of the  $k$ -means clustering algorithm to derive several clusters from the original dataset, the number being an input anonymization-related parameter, in a MapReduce parallel fashion. The algorithm runs in two rounds of MapReduce operations. The first round of MapReduce is where the mapper has the role of sorting the clusters based on their attribute value count (to form a key-value tuple where the key is an attribute value and the value is the attribute value count associated with the sensitive attribute). And the reducer is then responsible for deducing the sensitive attribute value count for each cluster.

In [Onesimu, Karthikeyan, Eunice, Pomplun, and Dang \(2022\)](#) the authors propose to anonymize both numerical and categorical attributes using attribute-based transformations. The transformation for numerical attributes involves calculating, for each Quasi-Identifier (numerical attribute), the *Interval Width* (IW) (consisting of the mean of the largest interval of values of the considered attribute), after which the algorithm builds the equivalence classes intervals that increment according to the IW from the smallest to largest attribute values. After building the ranges for the *Equivalence Classes* (EQs), initial tuple values are considered for mean calculation, leading to the anonymizing values that would replace the actual range of values of each EQ. In the case where attribute values, within one built EQ, are all equal, the algorithm

proceeds to alter the value to conceal the original attribute value. In the case of categorical attributes, they propose an anonymizing bucketization approach based on a clustering sub-method that uses a  $k$ -medoid approach to create clusters out of the dataset to anonymize and partition the clusters (assimilated to buckets) to generate  $l$ -diverse sensitive buckets. The  $l$ -diversity sensitive attribute values check (sensitive buckets generation) is based on the calculation of the probability distribution of the sensitive values within each sliced bucket, which should satisfy an  $l$ -proportioned threshold.

#### 4.6. Privacy via multidimensional anonymization in big data analytics systems

In this Section, we explore the concept of ensuring privacy via multidimensional anonymization in big data analytics systems by focusing on presenting some relevant state-of-the-art work.

[Xu et al. \(2006a\)](#) consists of a greedy anonymizing algorithms that maximize data utility and minimize a proposed metric called *Weighted Normalized Certainty Penalty* (NCP). Two approaches are proposed: the first one is the bottom-up approach, which starts by assigning each of the dataset tuples to a group and then tries to merge each of them (when the  $k$ -anonymity condition is not met) with another group and/or tuples (through a dataset scan pass) that would minimize the NCP metric. If the resulting merged group is larger than a certain threshold, then split it into smaller groups that satisfy the anonymization constraint related to the required group size ( $k$ ). Distances were used for numerical attribute values and hierarchies for categorical attribute values to compute the NCP metric between the groups and/or groups of tuples. The second one consists of a top-down approach where the original dataset is partitioned into consequent subsets while accounting for the minimization of the NCP metric, indeed, the partitions should be able to be further partitioned into smaller groups of tuples that minimize the NCP metric. At each iteration, partitions should verify the  $k$ -anonymity constraint. In the end, each group should satisfy  $k$ -anonymity, or an adjustment is performed.

[Li, Wong, Fu, and Pei \(2008\)](#) proposes an attribute hierarchical taxonomy approach for anonymizing the input dataset's attribute values through local recoding. The paper proposes a new metric, namely *Weighted Hierarchical Distance* (WHD), to assess the generalization results of the initial dataset. The metric measures the ratio of the sum of weights of each of the domain levels between the initial value and the generalized value and the sum of the weights of all levels of the hierarchical domain. A distortion between two tuples (the initial one and the generalized one) is then deduced by summing over all of the WHD distances between hierarchical domain levels of each of the attribute values of the initial and the generalized tuples. The distortion between tables is then the sum of the distortions of each tuple of the table. Distances between two tuples as well as between equivalence classes are also defined. The former is calculated by, first, identifying the *Closest Common Generalization* (CCG) between the tuples, and second, considering the sum of the distortions of the first tuple with CCG and the sum of the second with CCG. Finally, the distance between equivalence classes is defined as the equivalence class size weighted sum of the previous distance (tuples within an equivalence class being identical). The paper then defines the main algorithm named: *K-Anonymization by Clustering in Attribute hierarchies*, which loops over all of the equivalence classes (clusters) of an input dataset, and iterates over those who have a size lower than  $k$ , spots the closest equivalence class (cluster) to the previously mentioned one, and generalizes (local recoding generalization) both of the equivalence classes (the currently iterated and its closest one). The algorithm halts when no equivalence class remains with a size lower than  $k$ . To help the local recoding anonymization by clustering, the authors introduce the notions of stub and trunk, along with a splitting technique to delimit the data portions to be generalized or processed.

In Al-Zobbi, Shahrestani, and Ruan (2017), the authors introduce a multidimensional anonymization approach that adopts a Quasi-ID attribute-based splitting of the dataset before performing  $k$ -anonymity (lowering therefore heavy workloads related to anonymizing datasets with a large number of attributes). The idea is to consider for anonymization solely the attribute domains whose range is small, hence leading to improved information gain and efficiency. In order to implement this idea, they introduce a sensitivity-based mathematical expression that incorporates the notion of Quasi-ID attributes probabilities. The sensitivity value will dictate the degree of anonymization to be applied to the data (the minimum values of the respective Quasi-ID probabilities, or also the interval ranges to consider for anonymizing a Quasi-ID attribute). The probability relating to a Quasi-ID attribute (numerical) is nothing but the probability of drawing a number within the domain range of the attribute. For the categorical case, it is drawing a certain value from the set of attribute values of the considered Quasi-ID attribute. The sensitivity-based expression linear coefficient is based on these probabilities and on  $k$  parameter ( $k$ -anonymity). In addition to that, the sensitivity expression is linearly related to a user's data ownership level parameter, that inversely influences the sensitivity value: the higher that parameter is the lower the sensitivity value is and the higher the anonymity applied to the data. More specifically, in the previously described case, the range of domains considered for anonymization will be larger, thus impacting the raw data more heavily. In more detail, the algorithm they propose to anonymize the data works iteratively by considering at each iteration a subset of the Quasi-ID attributes to apply  $k$ -anonymity on. At each iteration, only Quasi-ID attributes that have the lowest probability scores (high attribute domain ranges) are considered for anonymization while grouping the records with the remainder of the Quasi-ID attributes. The implementation of this algorithm was achieved thanks to Apache Pig (Swarna & Ansari, 2017) which has permitted the definition of scripts necessary for achieving data record transformations.

Zhang, Qi et al. (2022) proposes an iterative version of the Mondrian based on a MapReduce approach. A cached indexing tree (containing basic information about partitions of data) named *Partition ID tree* (PID-tree) is maintained and shared across the system nodes. To tackle excessive computational overhead that bigger dataset may incur, they propose to apply serial MapReduce to each new level of partitions the driver instructions may yield after a new iteration is completed (rather than doing it recursively on each new partition). They propose an algorithm that accounts for two types of attributes: numerical and categorical. The driver "data splitting" iterations go on until all partitions satisfy a user-defined node's capability threshold (the maximum number of records that a node can fit in memory). Over the course of an iteration, the algorithm seeks to yield a set of partitions optimally split for anonymization (the anonymization is done in memory using MapReduce paradigm).

#### 4.7. Architectures and platforms for advanced privacy-preserving analytics on the clouds

In this Section, we examine some related work on architectures and platforms that enable advanced privacy-preserving analytics on the Clouds. Specifically, we delve into various frameworks and systems engineered to maintain data privacy while supporting sophisticated analytics within Cloud environment.

Zala et al. (2022) proposes a cloud-based e-health system enabling on-demand access to patients' EHRs (Electronic Health Records) while emphasizing the data privacy aspect. The proposed system architecture involves the use of third-party data centers to store data for later use as needed. To reach the end-users, data are conveyed over the network, where congestion and throughput issues are dealt with through data replication. The authors also propose a data acquisition model that allows fast and reliable data analysis that accounts for data processing time and priority. Besides, the data ingestion procedure accounts for the

frequency of consultations of the patient and their health emergency. Indeed, a window-based temporary information for acquiring data is put forward in order to selectively delimit most informative data to secure. The architecture proposed comprises: an access control module that enables data access to users based on their roles and privileges, a cryptographic encryption module that is responsible for concealing EHR information into a cover image (through AES-128 algorithm and Steganography's LSB technique) stored in a third-party data center, as well as a steganography module that ensures that decrypted data are within reach of the data access module. Finally, upon request from a user, the access control module validates the data through the anonymization module by hiding sensitive values and anonymizing Quasi-ID records before the data can be disclosed and delivered to the requesting user.

In Othman, Almalki, Chakraborty, and Sakli (2022), the authors propose a system to gather data from IoT sensors for medical care purposes that is energy consumption "friendly". The data reach a medical server through a bridging network enabled by WI-FI, Internet, or cellular networks. In this paper, the integrity of data is secured via *Homomorphic MAC* (H-MAC) so that no malicious attacker can alter the content of the data transiting over the network. In this case, the receiving side (medical server) will not require the decryption of the sent message in order to perform computation on it. The authors believe that this will provide further protection over the data and will shrink risky network-related exposure of the data. The authentication process between IoT sensing devices and the coordinator (data generating side), as well as the latter with the medical server are secured using traditional hash functions and secret-public key mechanisms. In addition, the confidentiality of the data is guaranteed using homomorphic encryptions (known for enabling computations over the encrypted data). In fact, the system aims to reduce energy consumption by reducing the data transmission rate within the network, which accounts for 70% of the total energy consumption (the rest is mostly attributed to the data aggregation-induced energy consumption). To accomplish that, they use DPM (Zhang, Deng Han and Zheng, 2018), an approach that is set to train prediction models both at sensing nodes and on the medical center side to inform on which data are most relevant for sending to the medical server. The core idea they leverage is that if data could be accurately enough (within the range of a defined threshold) predicted at the sensing node, then it could also be predicted at the medical server (since historical data used for training the models are the same for both sides). In addition, a simultaneous update of data-storing memories (needed for model learning) on both sides using predicted data values as well as newly sensed data values is executed. Provided this approach, it is possible to eventually differentiate useful from redundant data before sending them to the medical center. Eventually, pointless data transmissions will be avoided, thus engendering a reduction in network traffic and consequently a decrease in the total energy consumption of the system.

Abbasi and Mohammadi (2022) introduces a cloud system enabling organizations such as hospitals or medical centers to accumulate anonymized data from patients. Their system summarizes in a cloud system provider that does in-house anonymization for later dissemination of the concealed data. The anonymization approach they adopt consists of the superposition of several anonymization-enabling steps that, they argue, form an effective and efficient clinical data anonymization technique. The core anonymization workflow sums up in the refinement and standardization of the data through the use of a normal distribution as a reference to spot outlier records and discards them as they constitute the most vulnerable records to the attacks. Second, they employ  $k$ -means++ (Arthur & Vassilvitskii, 2006) in order to cluster the records in a way that, they argue, is beneficial to data utility. Indeed, they explain that with such an approach, the anonymization of the data (through  $k$ -anonymity) is made easier and more effective as it requires less generalization as data entries are adequately clustered (based on QID values). Finally, they apply the

$k$ -anonymity method to each cluster separately, yielding thereby the anonymized chunks of data that union to form the final anonymized dataset ready to be delivered to healthcare organizations.

In [Zhang et al. \(2015\)](#), a highlight of the limitations of the state-of-the-art anonymization techniques is put forward, indeed, the authors state that using standard anonymization techniques, sensitive information is still at risk of being disclosed. They add that if sensitive information of a QID class falls under a common semantic proximity domain, inference of such sensitive information is roughly possible through what they name as “categorical proximity breach”, where the sensitive information is roughly determined through its potential semantic proximity domain belonging. In order to implement their idea, they propose a distance metric over sensitive attributes. Mathematically, the metric is defined as the distance between two sensitive values over the longest possible distance between two sensitive values that are defined by the taxonomy tree of a given attribute. The distance between two sensitive values of a certain attribute is calculated through their lowest common ancestor. According to their framework, the anonymity of sensitive values is then ensured by setting a threshold level of dissimilarity. In addition to that, they define a proximity index over two sensitive values that are based on their distance, as earlier mentioned, and extend that definition to QID groups. Furthermore, a distortion metric is defined to capture the data distortion resulting from the generalization process. Through the mentioned definitions, they define a proximity-aware clustering approach that aims at minimizing both the proximity of clusters and the distortion resulting from the generalization process. To further refine their approach, they built a scalable version of the previously mentioned algorithm. Indeed, they define a two-phase proximity-aware clustering using MapReduce. The first step of the process generates the clusters through  $t$ -ancestors method, where the clusters are formed by grouping records together with their nearest ancestor. The ancestor of the cluster is then recomputed accounting, for the new formation: the computation of an ancestor record of a cluster is performed by calculating the median of QID attribute values of the cluster (numerical values) or the lowest common ancestor of the same QID attribute values of the cluster records (categorical values). The second step is about applying an agglomerative clustering method to each of the generated clusters to yield a set of clusters from each initial cluster. Generalization is then performed on each of the clusters resulting from the second phase. The agglomerative clustering algorithm idea is the merging of the closest clusters together while considering the distance between the clusters to be a function of the farthest two records belonging to each cluster (e.g., the maximum value of the proximity distances between records of the considered clusters).

## 5. Definition of main properties

In this Section, we define a number of methods and/or techniques related to data anonymization, data analytics, and more broadly, data mining that will enable a later categorization of the previously described literature research works. Basically, this consists of defining the *main properties* of the literature investigated in this study.

### 5.1. Data utility

*Utility* (data) refers to the usefulness of the data for analysis, usually after anonymization. To capture the utility of the data, metric expressions may involve calculations over both the final anonymized data and the initial data. More specifically, a data utility measure should capture the difference between the distributions of data before and after the anonymization. Two notable aspects when assessing data quality through a utility metric are information loss resulting from anonymization and attribute importance. A well-defined utility metric should be able to capture intrinsic aspects impacting data quality for the considered application. For the sake of clarity, and since amalgamation is often made between data utility and data quality, we

emphasize that the two notions are different. While data utility is useful to gauge before applying data mining techniques, data quality applies in a broader context. Indeed, data quality concerns features related to the data itself (as opposed to its usefulness for another process), such as data accuracy (for example, after a data sanitation process, it evaluates how different an attribute value is from its initial raw value), data completeness (evaluates the degree of missed data after sanitation), and data consistency (being the degree of attribute correlation holding after sanitation).

### 5.2. Data loss techniques

A *data loss* technique refers to a data processing technique that results in a partial data waste posterior to a “lossy” task (in particular, an anonymizing procedure). For example [Wang, Du, Luo, and Li \(2019\)](#) is an anonymizing model designed to deal with transactional data streams. The model uses data normalization and suppression to anonymize sensitive values. Unfortunately, these techniques are known to cause data loss. Another worth-mentioning research work is described in [Wong et al. \(2007\)](#), where an extended version of  $(\alpha, k)$ -anonymity is defined. The suggested method involves applying  $(\alpha, k)$ -anonymity and generating two tables to be privately published. The two tables share a group-ID column that assigns to each tuple an equivalence class identifier. The paper demonstrates how the method could improve data distortion compared with traditional data publishing (one generalized table). In addition, the authors argue that the original data cannot be recovered unless a lossy join is performed on the generated tables.

### 5.3. Data reconstruction

*Data reconstruction* refers to the set of algorithms that attempt to recover initial data that underwent a lossy process. As their name suggests, such processes incur a loss of data, so the main challenge of data reconstruction methods is to ensure that the entirety of the initial data is recovered (lossless reconstruction). Examples of data reconstruction methods are numerous in the literature, we mention ([Qin et al., 2022](#)), where compressed traffic data are reconstructed in the context of a large-scale Internet of Vehicles using a neural network approach. Some researches are focused on a specific application of data reconstruction, as for example, for remote sensing images ([Zhang, Yuan, Zeng, Li and Wei, 2018](#)) or for recovering data in multidimensional time series ([Baraldi, Maio, Genini, & Zio, 2015](#)), while others are more general and apply to any type of data as [Intawichai and Chaturantabut \(2022\)](#) where a method is based on a least-squares approximation in combination with a randomized *Signal Value Decomposition* (SVD) technique is proposed. In addition, certain other research works tackle data reconstruction in the field of privacy-preservation: for instance, [ElSalamouny and Palamidessi \(2022\)](#) describes a distribution recovery method for anonymized data that attempts to recover an optimal utility for users’ sensitive data. Another worth noticing research work is [Jing, Miao, Song, and Chen \(2019\)](#), which provides a solution to reconstruct data after applying a location-protecting differential privacy anonymizing method in the context of an edge network.

### 5.4. Data lossless techniques

A *data lossless* technique refers to a data processing technique that keeps data intact through the considered data process. Literature research works that tackled this challenge are numerous, among them, we mention: [Navale and Mali \(2019\)](#), where a lossless anonymizing technique for association rules is detailed. The proposed technique aims at protecting sensitive rules that are accessed from an association rules database using the *Firefly* optimization algorithm ([Yang, 2010](#)). Another research that dealt with data lossless techniques is described in [Shailaja and Rao \(2022\)](#), where the proposed algorithm

generates optimal keys for data sanitation as well as for data restoration through a combination of methods, namely, the particle swarm optimization (Wang, Tan, & Liu, 2018) and the whale optimization algorithm (Mirjalili & Lewis, 2016). Another interesting research work (Li & Kot, 2010) proposes a fingerprint database anonymization through a novel technique defining a “pixel embeddability criterion” that decides whether or not a certain pixel in the fingerprint image could be highlighted as a hiding element of the image. Using a pixel relationship-related expression, based on logical “NOT” and “AND”, the initial fingerprint is then recovered through a pattern reconstruction process.

### 5.5. Anonymization

*Anonymization* is the process of concealing all key identifying features that may expose sensitive information about individuals when data owner instances (hospitals, insurance companies, etc.) share their data with a third party (clinics, research centers, etc.). Its end goal is to protect sensitive information (e.g., about patients in the healthcare domain) from being disclosed to an untrusted party by depersonalizing data records. Indeed, anonymizing data records is to preserve privacy and individuals information. Typically, a data collection is divided feature-wise into *QID* (Quasi-Identifier) attribute values and sensitive values. The anonymization process consists in either suppressing, generalizing, perturbing, permuting, or anatomizing (bucketing) the *QID* attribute values. Suppression happens when attribute values are replaced by an asterisk, generalization is achieved through replacing actual values with a generalized value from their hierarchy line (the generalization is based on a priorly defined generalization attribute values hierarchy), perturbation is done through adding noisy data to the records to anonymize in a calculated way as to keep certain statistical characteristics about the records unaltered, permuting is achieved by partitioning the records into groups where sensitive values are shuffled within each group, finally anatomization (or anatomy or bucketization) is realized through decoupling quasi-identifiers and sensitive attributes in two separate tables while maintaining linking information as to make sensitive values ascribable to a given quasi-identifier tuple. Anatomization is often applied in combination with either *k*-anonymity or *l*-diversity.

### 5.6. Privacy-preservation

*Privacy-preservation* needs stem from data confidentiality requirements that aim to protect the privacy of patients. These requirements are usually imposed by national and international data protection laws such as *US Health Insurance Portability and Accountability Act* (HIPAA) (Centers for Medicare & Medicaid Services, 1996) and the (GDPR). Privacy-Preservation is established through a set of techniques and approaches aimed at transforming sensitive data into an unrecoverable format. Most often, privacy-preservation is tied to the notion of data sharing. Indeed, data sharing is the practice of permitting a third-party to acquire the data to generate new insights and eventually enable fine-grained personalized medicine. An important application in the healthcare area is anonymization, whose goal is to obfuscate the identity of individuals/patients to the greatest extent possible while keeping bonds with their sensitive information. To note that sensitive information should be minimally suppressed for data utility to be maintained. Examples of privacy-preserving methods include: *k*-anonymity (Sweeney, 2002) and *l*-diversity (Machanavajjhala, Kifer, Gehrke, & Venkatasubramanian, 2007).

### 5.7. Data masking

*Data masking* encompasses a set of techniques responsible for obscuring data in a matter of privacy-preservation. Contrary to other anonymizing techniques, data masking aims at concealing the data whilst keeping the initial structure of the attribute values intact using

techniques such as *Shuffling* (Muralidhar & Sarathy, 2006) changes the order of attribute values within a dataset column, *Nulling* (using generic symbol instead of the actual letters/numbers), *Substitution* uses a different synthetic (fake) attribute value to mask a target data value, *Randomization* (Cadar, Akritidis, Costa, Martin, & Castro, 2008) consisting of replacing attribute values with random data, and also *Skewing*, which acts on the attribute values through domain distribution variance to alter their initial authentic values. *Blurring* is also a data masking method that aims to alter each individual attribute value accordingly to yield updated values falling under a pre-defined range. A less common method is *Averaging*, where a summarizing aggregate relationship to the real data is maintained while certain attribute values undergo updates.

### 5.8. Perturbation

*Perturbation* is a technique used to alter an attribute value for the purpose of anonymizing it. Typically, this technique leverages the use of a known distribution (e.g., the distribution of the data to anonymize) to alter the initial data, hence representing them in a privacy-protecting form (probability distribution-based perturbation). On the other hand, perturbation could be done in a “random” way using synthetically generated data to replace the initial values with new ones in a once for all manner (fixed data perturbation). In both types of perturbations, noisy data are added to the original data in such an ideal way to preserve certain statistical knowledge engraved in the data, such as the domain mean, variance, or correlation. Unfortunately, many literature perturbation techniques introduce bias (defined as the difference in query responses using perturbed and original data) when applied to the data, leading to the wiping of certain statistical information (summary measures) of the data, such as the mean, as well as the obliteration of correlations between the attribute values. Note that one perturbation technique that withstands all four types of biases (Muralidhar, Parsa, & Sarathy, 1999) is the *General Additive Data Perturbation* (GADP) (Muralidhar et al., 1999).

### 5.9. Bucketization

*Bucketization* is an anonymization process that decouples *equivalence classes* (*QID* attribute values in record groups) from sensitive attributes yet keeps the link between the two for data analysis purposes. By loosening the bonds between the *QID* and the sensitive attributes, sensitive attributes within a bucket may be equally attributed to a given record (since they belong to a bucket and not to an ordered list anymore). Technically, the *QID* attributes are de-associated from the sensitive attributes, hence resulting in the creation of two tables connected through a group membership column *Group-ID*. The *Group-ID* column is responsible for “roughly” keeping track of the origin of sensitive values and to which group of records they belong. A count of the sensitive values within each bucket is also maintained. Note that the count of different IDs in the *Group-ID* column represents the total number of generated buckets. This may lead to strengthening privacy and lowering data attack risks. An example of such methods is described in Cao, Karras, Kalnis, and Tan (2011).

### 5.10. Local recoding

*Recoding* data means applying a privacy-preserving transformation over it (e.g., generalization). *Locally recoding* data is to permit one attribute value in a *QID* equivalence class (Quasi-ID) to have two or more transformed (e.g., generalizing) values. In other words, local recoding of a domain of values means the possibility of assigning different privacy-protecting (e.g., generalizing) values to the domain of values in a flexible way. The mapping of a domain of values to its generalized counterpart is performed on a tuple-basis. Indeed, attribute values are transformed (e.g., generalized) “by tuple” (and not by domain). More specifically, to locally recode a domain of values, a “neighboring”

portion of occurrences will be considered for a certain generalization, while other “close” occurrence groups will be considered for other generalizing values. This flexibility in dealing with the domain values has the advantage of potentially securing more data utility by minimizing information loss during the anonymization process. Some notable examples of anonymization procedures leveraging local recoding are described in Aggarwal et al. (2010) and Xu et al. (2006b).

### 5.11. Global recoding

*Global recoding* domain values during an anonymization is to generalize all of occurrences of an attribute domain to a one generalizing value in order to reach the targeted anonymization threshold. Note that such methods may overgeneralize the domain values, leading to an increased data loss. The information loss resulting from the global recoding is much more important than of the local recoding. Additionally, global recoding could be considered a special case of local recoding.

### 5.12. Data encryption

*Data encryption* describes a set of techniques employed to encode data, typically using a public/private key encryption algorithm, prior to manipulating them (e.g., sending them over a network). Through a data encryption process, plain data are transformed into ciphered data. The other way round, the decryption process turns the ciphered messages into plain text. The core idea is to use iterations (called *rounds*) of text *substitution* and *permutation* on the message to encrypt, where bits are accordingly altered. An example of such techniques is described in Balilo, Gerardo, and Byun (2019).

### 5.13. Attribute-focused techniques

*Attribute-focused techniques* aim to anonymize attribute values by performing attribute-related operations. Typically, this type of technique differentiates the anonymization procedure to apply depending on the type of the attribute (categorical or numerical). In both cases, the attribute-focused technique will enable the anonymization of the attribute values using two different approaches (one for numerical attribute values and another for categorical attribute values). Examples include interval width identification to group numerical values and the use of the mean for replacement of the target values. For categorical attributes, examples include an id-based matching technique (Majeed, 2019) or clustering (of tuples through PAM k-medoid algorithm Park & Jun, 2009) followed by tuple partitioning (where bucketed tuples satisfy an anonymization check, e.g., *l*-diversity) (Onesimu et al., 2022).

### 5.14. Data-range-focused techniques

Contrary to attribute-focused techniques, *data-range-focused techniques* focus the attention on obtaining anonymity in target datasets via applying specific data processing procedures over ad-hoc collections of data items (stored in those datasets). The main idea here consists in devising models and techniques that intelligently exploit specific properties of data (e.g., distributions, ranges, maximum value, minimum value, and so forth) to achieve the desired anonymization. For instance, this is the case of *aggregation-based anonymization* (e.g., Benkaouz, Erradi, and Freisleben (2015)), or *obfuscation-based anonymization* (e.g., Ardagna, Cremonini, Damiani, di Vimercati, and Samarati (2007)). Some approaches also combine *multiple* methods to obtain this effect (e.g., Ren et al. (2021)).

### 5.15. Error-based metrics

*Error-based metrics* apply particular criteria that are defined starting from a given *error metrics*, which, in turn, can be native or derived from multiple (error) metrics. For instance, in the case of OLAP data cubes, a popular error-based metric consists of minimizing the *query answer error* (e.g., Cuzzocrea and Serafino (2009)), given a population of “typical” queries applied to the target data cube. Other initiatives make use of *encryption techniques* based on the maximization of decryption errors, in order to achieve the security-based privacy of target records (e.g., Abdulkadir, Waziri, Alhassan, and Ismaila (2022)), for instance in the healthcare setting. Furthermore, other approaches consider the error an hypothetical adversary could make in detecting dataset properties (like, for instance, the distributions of specific attributes in the target dataset), as in Hoh and Gruteser (2005).

### 5.16. Query-based metrics

*Query-based metrics* are a type of metric based on answers by comparing those prior and posterior to the application of a transforming technique (e.g., anonymization, compression, etc.) on target data. This is important in order to inform users about the quality of transformed data and whether or not it fulfills a certain quality requirement. This metric may also be useful to quantify the data alteration caused by the applied transformation. For example, in the field of OLAP, query-based metrics are an assessment of the impact of a data transformation method (anonymization, compression, etc.) on the data utility/data loss (or information loss) balance. Further on this, OLAP query-based metrics could then evaluate such impact on the data by comparing the exact answer and the approximate answer (those submitted on the original data and the transformed data, respectively).

### 5.17. Information loss metrics

*Information loss metrics* are used to evaluate the degree of lost information during the anonymization process. More specifically, the metric evaluates the proportion of lost information due to anonymizing a certain domain. In the case of generalization, for example, it can be expressed as the max and min bound differences of a generalized domain over the same bound difference of the same domain. Whereas, in the case of suppression, the information loss metric can be expressed through the proportion of suppressed record count over the total record count for a domain. In addition, in the case of perturbation, information loss is measured through dissimilarity between the initial and the sanitized data. We mention, for the sake of an example of such metrics, dissimilarity (Bertino, Fovino, & Provenza, 2005), the generalized information loss metric (Nergiz & Clifton, 2007), or also the discernibility metric (Bayardo & Agrawal, 2005).

### 5.18. Utility-based metrics

*Utility-based metrics* measure the impact of anonymization on the utility of the data by considering the information loss for each attribute as well as their importance for data analysis. Differently from an information loss metric, a utility-based metric incorporates attribute weighting with respect to their utility for data analysis. For instance, we mention “*Distortions of Generalization of Tuples*” (Li, Wong, Fu, & Pei, 2006), “*Weighted Normalized Certainty Penalty*” (Xu et al., 2006a) or also “*Total Information Loss*” metric (Byun, Kamra, Bertino, & Li, 2007). Other notable examples of data utility metrics are described in Karr, Kohnen, Oganian, Reiter and Sanil (2006) and Karr, Oganian, Reiter and Woo (2006).

### 5.19. Aggregation-based methods

An *aggregation method* uses data collected from multiple sources in order to perform certain aggregation operations on them. In the field of privacy-preservation, aggregation may be performed over encrypted data in order to keep data safe while completing the aggregation task. The technique allowing such private aggregations is called homomorphic encryption. The aggregation is used to conceal the original data while enabling their later manipulation by the components of the anonymizing system. At the end of the process, aggregated data will be delivered to the end user in order to enable them to extract insights from the statistical properties that are within these aggregations. Method examples include the use of complex aggregations over encrypted data (Liu et al., 2023).

### 5.20. High-dimensional data

*High-dimensional data* are data that are characterized by an elevated number of features (or attributes), which is mostly the case in the healthcare domain. More broadly speaking, for a certain collected observation, the number of observed variables would be very high. For example, in the clinical data, if patient records are being the considered observations, then the number of record characterizing features (observed variables) is particularly high. This kind of data raises a lot of concerns about whether or not they could be effectively and efficiently handled (and most importantly anonymized), as this is remaining an open challenge nowadays commonly known as the curse of dimensionality. For example, it was suggested (Aggarwal, 2005) that in order to privacy-protect high dimensional data it is necessary to suppress a high number of attributes, thus negatively impact the data utility and hampering a productive analysis of the data.

### 5.21. OLAP queries

*OLAP queries* are requests submitted to *OLAP data cubes* that involve aggregations over designated target cells for the purpose of analyzing the underlying OLAP cubes. Most often, these queries are directed towards special servers known as *Data Warehouses* where the OLAP data cubes reside. We note important types of OLAP queries named range queries, where the aggregation operation is applied over targeted cells (containing the target measures) enclosed within the specified delimiting numerical ranges. For instance, the range-SUM family of queries computes the sum of the measures associated with cells within the specified boundaries. Another commonly used operation is the range-COUNT family of queries, which counts the number of the specified range-enclosed cells.

### 5.22. Centralized settings

These settings refer to a centralized environment where algorithms or systems leverage a centralized approach for storing and processing (e.g., data). All of the required operations are then done within the same location. Anonymizing algorithms leveraging such settings perform tasks on a single machine in a single location over an entire dataset (e.g., Machanavajjhala et al. (2007) and Sweeney (2002)).

### 5.23. Distributed settings

These settings refer to a distributed environment, where algorithms or systems leverage distributed capabilities for storage and computation. Usually incurs the use of a network to transport data or computed values to two geographically distant locations (e.g., client-server setting). In the case of data anonymization, distributed components may need to operate in parallel in an independent man-

ner, where the anonymization task is shared across the participating “nodes” (often called “workers”) in such a way that benefits the performance (e.g., processing time) while maintaining a good quality for the rendered anonymized dataset (e.g., in terms of utility). Many research works have dealt with distributed settings for anonymizing large datasets, for example, in healthcare (Mohammed et al., 2010) or more broadly for big data (di Vimercati et al., 2023).

### 5.24. Scalability

*Scalability* refers to a system’s capability to cope with a larger workload by dynamically, adequately, and automatically self-allotting sufficiently needed resources (storage and computation) to function properly. A common heuristic that scalability-enabling algorithms may adopt, is to lower data loads to transfer over a network by considering a simplified representation of the considered data. A fact worth noting is that several literature works enable scalability through distributed architectures leveraging the MapReduce programming paradigm over the cloud, such as in Mehta and Rao (2022). Another example of scalable systems is described in di Vimercati et al. (2023), where a Mondrian-based anonymizing method was defined by extending the *Mondrian algorithm* (LeFevre, DeWitt, & Ramakrishnan, 2006) through an innovative quantile-based partitioning approach leveraging rankings of the dataset attribute values. In their approach, worker nodes are assigned exclusive portions of the dataset optimizing thereby data exchange rates between the nodes, which then can independently anonymize their respective local data fragments. The data portions are defined using a coordinator, which sets, accordingly, the conditions of the partitioning.

### 5.25. Multidimensional big data analytics model

*Multidimensional big data analytics model* is a scheme enabling the multidimensional analysis of data by being aware of the underlying data structure. A considerable number of research works have defined multidimensional analytics models, such as in Chen (2021) and Shi (2021). A well-defined analytics model would be conceived based on the type of analysis sought and the application area. A well-known multidimensional data analytics-enabling tool is OLAP cubes. Indeed, an OLAP cube permits the multidimensional representation of data in a pre-analysis phase. Typically, OLAP is leveraged to build a suitable model for a target analysis application. For instance, Doreswamy and Harishkumar (2018) explicitly defines a multidimensional ROLAP data cube model and specifies star and snowflake schemas for air pollution data to analyze them and eventually enable decision-making and knowledge discovery. Another example is described in Saidi, Trabelsi, and Ghézala (2018), where a terrorist data warehouse model is proposed. Indeed, a galaxy schema is defined to ease the investigation of terrorist acts for the specialists and to enable decision-making. The system also permits MDX querying over terrorist attack data to enable results depending on the investigator’s intuitions. Furthermore, a privacy-preserving relating data analytics model is described in Cuzzocrea, Leung, Olawoyin, and Fadda (2021) which applies to the case of temporal open big data.

## 6. Survey summary table

In this Section, we introduce Table 1, where we summarize a selection of references according to their membership to the before-described properties. This table can be interpreted as one of the most noticeable results of our study.

**Table 1**  
Reference classification by property.

| Property                                  | Reference           |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
|-------------------------------------------|---------------------|-----------------------|-------------------|------------------|------------------------|-----------------------|-------------------|-------------------------|------------------------|--------------------------|-----------------------|-----------------------|------------------------|-----------------------------|--------------------|----------------------|-------------------|------------------------------|----------------------------|---------------------|--------------------|-------------------|------------------------|--------------------------------|----------------|
|                                           | Zhang et al. (2015) | Prasser et al. (2016) | Xu et al. (2006a) | Li et al. (2008) | Abdalaal et al. (2013) | Ghinita et al. (2011) | Liu et al. (2023) | Zhang, Qi et al. (2022) | Tanaglia et al. (2021) | Bochicchio et al. (2016) | Onesimu et al. (2022) | Onesimu et al. (2021) | Mohammed et al. (2010) | Abbasi and Mohammadi (2022) | Zala et al. (2022) | Othman et al. (2022) | Kim et al. (2017) | Cuzzocrea and Bertino (2014) | Cuzzocrea and Sacca (2012) | Zheng et al. (2022) | Peng et al. (2022) | Liu et al. (2022) | Al-Zobbi et al. (2017) | Lawrance and Jesudhasan (2021) | Sweeney (2002) |
|                                           |                     |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Data utility                              |                     | ✓                     | ✓                 | ✓                | ✓                      |                       |                   |                         |                        |                          |                       |                       | ✓                      |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Data loss technique                       | ✓                   |                       | ✓                 | ✓                | ✓                      |                       | ✓                 |                         | ✓                      |                          | ✓                     |                       | ✓                      |                             |                    |                      | ✓                 |                              | ✓                          |                     |                    |                   |                        |                                |                |
| Data reconstruction                       |                     |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   | ✓                            |                            |                     |                    |                   |                        |                                |                |
| Data lossless technique                   |                     | ✓                     |                   |                  |                        | ✓                     |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Anonymization                             | ✓                   | ✓                     | ✓                 | ✓                | ✓                      | ✓                     | ✓                 | ✓                       |                        |                          |                       | ✓                     |                        | ✓                           |                    | ✓                    | ✓                 | ✓                            | ✓                          | ✓                   | ✓                  | ✓                 | ✓                      | ✓                              | ✓              |
| Privacy-preservation                      | ✓                   |                       | ✓                 | ✓                | ✓                      | ✓                     | ✓                 | ✓                       | ✓                      |                          | ✓                     | ✓                     | ✓                      | ✓                           | ✓                  | ✓                    | ✓                 | ✓                            | ✓                          | ✓                   | ✓                  | ✓                 | ✓                      | ✓                              | ✓              |
| Data masking                              |                     |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Perturbation                              |                     |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Bucketization                             |                     |                       |                   |                  |                        |                       |                   |                         |                        | ✓                        |                       |                       |                        |                             |                    |                      | ✓                 |                              |                            |                     |                    |                   |                        |                                |                |
| Local recoding                            | ✓                   |                       | ✓                 | ✓                |                        |                       |                   |                         |                        | ✓                        |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Global recoding                           |                     | ✓                     |                   |                  | ✓                      |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      | ✓                 |                              |                            |                     |                    |                   |                        |                                |                |
| Data encryption                           |                     |                       |                   |                  |                        |                       | ✓                 |                         |                        |                          |                       |                       |                        |                             | ✓                  | ✓                    |                   |                              | ✓                          | ✓                   | ✓                  |                   |                        |                                |                |
| Attribute-focused technique               |                     |                       |                   | ✓                |                        | ✓                     | ✓                 |                         |                        |                          |                       |                       | ✓                      |                             | ✓                  | ✓                    |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Data-range-focused techniques             |                     |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Error-based metrics                       |                     |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Query-based metrics                       |                     |                       |                   |                  |                        |                       |                   |                         |                        |                          |                       |                       |                        |                             |                    |                      |                   | ✓                            |                            |                     |                    |                   |                        |                                |                |
| Information loss metrics                  | ✓                   |                       |                   |                  |                        |                       |                   | ✓                       |                        |                          |                       | ✓                     | ✓                      | ✓                           |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Utility-based metrics                     |                     |                       | ✓                 | ✓                |                        | ✓                     |                   | ✓                       |                        |                          | ✓                     | ✓                     | ✓                      | ✓                           |                    |                      |                   |                              |                            |                     |                    |                   |                        |                                |                |
| Aggregation-based technique               |                     |                       |                   |                  |                        |                       | ✓                 |                         |                        |                          | ✓                     |                       | ✓                      |                             |                    |                      |                   | ✓                            | ✓                          |                     |                    | ✓                 |                        |                                |                |
| High-dimensional data                     | ✓                   | ✓                     | ✓                 | ✓                | ✓                      | ✓                     | ✓                 | ✓                       | ✓                      | ✓                        |                       | ✓                     | ✓                      |                             | ✓                  |                      |                   |                              | ✓                          |                     |                    | ✓                 | ✓                      |                                |                |
| OLAP queries                              |                     |                       |                   |                  |                        |                       |                   |                         | ✓                      | ✓                        | ✓                     |                       | ✓                      |                             |                    |                      | ✓                 | ✓                            | ✓                          | ✓                   |                    |                   |                        |                                |                |
| Centralized setting                       |                     | ✓                     | ✓                 | ✓                | ✓                      | ✓                     | ✓                 | ✓                       | ✓                      | ✓                        |                       |                       |                        |                             |                    |                      |                   | ✓                            | ✓                          | ✓                   |                    |                   |                        | ✓                              | ✓              |
| Distributed setting                       | ✓                   |                       |                   |                  |                        |                       | ✓                 |                         | ✓                      | ✓                        |                       | ✓                     |                        | ✓                           | ✓                  | ✓                    |                   | ✓                            | ✓                          | ✓                   |                    | ✓                 | ✓                      | ✓                              | ✓              |
| Scalability                               | ✓                   |                       |                   |                  |                        | ✓                     | ✓                 | ✓                       | ✓                      | ✓                        |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    | ✓                 | ✓                      | ✓                              |                |
| Multidimensional big data analytics model |                     |                       |                   |                  |                        |                       | ✓                 | ✓                       | ✓                      | ✓                        |                       |                       |                        |                             |                    |                      |                   |                              |                            |                     |                    | ✓                 |                        |                                |                |

## 7. Challenges and future research directions

By analyzing active literature, several challenges and future research directions in the context of privacy-preserving multidimensional big data analytics arise. In the following, we report on some of the most noticeable ones.

- 1. Large-Scale Big Multidimensional Data.** One critical challenge of the investigated research area consists of how to make privacy-preserving multidimensional big data analytics techniques and algorithms *scalable* while the volume of underlying multidimensional big data repositories grows (e.g., [Niu et al. \(2018\)](#)). This is a very critical challenge that will deserve a great deal of attention in the future years.
- 2. Maintaining Privacy of Big Multidimensional Data.** When the privacy of underlying multidimensional big data repositories is achieved, how to *maintain* this privacy under rapidly-changing data? As widely known, a typical characteristic of multidimensional big datasets is represented by the fact that they are inherently evolving data, and this may easily impact on the validation of *actual* privacy constraints over datasets (e.g., [Derbeko, Dolev, and Gudes \(2018\)](#)).
- 3. Provenance of Multidimensional Big Data.** *Provenance* refers to detecting and analyzing the origin and lineage of data. When applied to multidimensional big datasets (e.g., [Hu et al. \(2020\)](#)), this problem becomes critical since these data are complex and very heterogeneous in nature. Therefore, provenance of multidimensional big data represents a harder case to investigate, and it will play a leading role in future big data research.
- 4. Uncertain Multidimensional Big Data.** *Uncertainty* is a typical feature of multidimensional big data, due to the fact that

such data are subjected to errors of various natures (including coding errors, transmission errors, human errors, and so forth). How privacy-preserving multidimensional big data analytics techniques and algorithms deal with such uncertainty in data (e.g., [Huang, Wang, and Chawla \(2020\)](#))? The latter is a leading question to be addressed by future research efforts.

- 5. Integration with Modern Big Data Processing Platforms.** Last but not least, the success and, moreover, the standardization of privacy-preserving multidimensional big data analytics techniques and algorithms will also depend on their capability of being integrated with modern big data processing platforms, like *Hadoop* and *Spark*. This is another future key point of the investigated research context.

## 8. Conclusions and future work

In this paper, we have presented and critically discussed a comprehensive survey on privacy-preserving multidimensional big data analytics models, methods, and techniques over big data. We identified seven specialized topics within this broader scientific context, and according to these pillars, we developed our study. A very detailed related work analysis of state-of-the-art proposals, classified by means of the previous pillars, has been provided. In addition to this, a review of future research challenges in the investigated research domain has been illustrated.

For what specifically regards the research questions provided in Section 1.1, we can conclude that our paper has finally achieved the following answers (for the sake of clarity, here we report the research questions we introduced above):

- **RQ1.** Which are the main pillars of the research corpus in privacy-preserving multidimensional big data analytics?

**AQ1.** The research question **RQ1** has been answered via the identification of seven pillars in the context of privacy-preserving multidimensional big data analytics (see Section 3) and the classification of relevant state-of-the-art proposals with respect to these pillars (see Section 4), thus defining a systematic approach to deal with the investigated scientific context.

- **RQ2.** Which are the most-relevant and emerging proposals in the context of privacy-preserving multidimensional big data analytics, and how to well-summarize them?

**AQ2.** The research question **RQ2** has been answered via the identification of the main properties of the literature corpus in the context of privacy-preserving multidimensional big data analytics (see Section 5) and the introduction of a proper summary table about the most-relevant and emerging proposals with respect to these main properties (see Section 6).

- **RQ3.** Which are the most-noticeable and emerging challenges and future research directions in the context of privacy-preserving multidimensional big data analytics?

**AQ3.** The research question **RQ3** has been answered via the identification, definition and discussion of several challenges and future research directions in the context of privacy-preserving multidimensional big data analytics (see Section 7), thus opening the door to future research efforts in the investigated scientific context.

Future work is oriented towards extending our study on other aspects of the more general big data processing process, beyond to privacy-preserving big data analytics like addressed in this study, such as *machine learning over big data* (e.g., Leung, Braun, Hoi, Souza, and Cuzzocrea (2019) and Leung, Chen, Hoi, Shang and Cuzzocrea (2020)) and *big data visualization* (e.g., Barkwell et al. (2018) and Leung, Chen, Hoi, Shang, Wen et al. (2020)), still in a privacy-preserving manner.

#### CRedit authorship contribution statement

**Alfredo Cuzzocrea:** Conceptualization, Methodology, Validation, Resources, Writing – original draft, Writing – review & editing. **Selim Soufargi:** Validation, Writing – original draft.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Data availability

No data was used for the research described in the article.

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